

ALEXANDER RYABCHENKO

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PROFILE

PhD student at the University of Toronto and Vector Institute researching the statistical and computational foundations of modern machine learning, including reinforcement learning, optimization, and post-training.

EDUCATION

Doctor of Philosophy (PhD), Statistical Sciences <i>University of Toronto</i>	2026 – Present Toronto, ON
• Advisors: Murat A. Erdogdu, Wenlong Mou, Stanislav Volgushev.	
Honours Bachelor of Science (HBS), Mathematics and Computer Science <i>University of Toronto</i>	2021 – 2025 Toronto, ON
• Cumulative GPA: 4.00/4.00 (Average grade: 98.0%).	
• Governor General's Silver Medal (highest academic standing), John Black Aird Scholarship (top student).	

PUBLICATIONS

- **A.R.**, Jian Qian, Wenlong Mou. Adaptively Incorporating Directional Hints into Zeroth-Order Optimization. To appear in *Advances in Neural Information Processing Systems* (NeurIPS 2026). [arXiv:2609.08277](#).
- **A.R.**, Idan Attias, Daniel M. Roy. Capacity-Constrained Online Convex Optimization with Delayed Feedback. To appear in *NeurIPS 2026*. [arXiv:2606.11711](#).
- **A.R.**, Idan Attias, Daniel M. Roy. A Reduction from Delayed to Immediate Feedback for Online Convex Optimization with Improved Guarantees. To appear in *NeurIPS 2026*. [arXiv:2602.02634](#).
- **A.R.**, Wenlong Mou. Reinforcement Learning with Action-Triggered Observations. In the *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*. [arXiv:2510.02149](#).
- **A.R.**, Idan Attias, Daniel M. Roy. Capacity-Constrained Online Learning with Delays: Scheduling Frameworks and Regret Trade-offs. In the *Proceedings of the 38th Conference on Learning Theory (COLT 2025)*, pp. 4973-5014. [arXiv:2503.19856](#).

PREPRINTS

- Idan Attias^{*}, Orin Levy^{*}, **A.R.**^{*}, Yishay Mansour, Uri Stemmer. Vanilla Policy Optimization Is Both Optimal and Differentially Private for Stochastic Contextual Bandits. [arXiv:2609.33888](#).

^{*}Equal contribution.

AWARDS

Connaught International Doctoral Scholarship (50,000 CAD)	2026 – 2031
Governor General's Silver Medal	2025
• Canada's most prestigious academic award, established in 1873; awarded to the undergraduate student with the highest academic standing upon graduation.	
John Black Aird Scholarship	2025
• Award recognizing the top graduating student at the University of Toronto.	
E. A. Robinson Medal in Sciences	2025
• Medal recognizing the top graduating science student at the University of Toronto.	
Vector Institute Research Grants (30,000 CAD)	2023 – 2024
University of Toronto International Scholar Award (180,000 CAD)	2021 – 2025
Undergraduate Scholarships & Research Awards at the University of Toronto	2021 – 2025
• MCS Research Award (7,500 CAD)	
• Joseph Alfred Whealy Scholarship (6,000 CAD total; four-time recipient)	
• Peter Fantham Memorial Scholarship in Mathematics (2,000 CAD)	
• Mariano A. Elia Scholarship (1,500 CAD total; two-time recipient)	
• Doris Geddes Scholarship in Mathematics (1,000 CAD)	

RESEARCH EXPERIENCE

Graduate Student Researcher

2023 – Present

Vector Institute

Toronto, ON

- Working on the theoretical foundations of modern AI: LLM post-training, scaling laws, optimization.
- Developed theory for sequential decision-making under delayed and partial feedback, including a novel framework for learning under capacity constraints, motivated by real-world applications such as online A/B testing in recommender systems and clinical trials. The resulting work, conducted under the supervision of Professor Daniel M. Roy, was published and presented at COLT 2025.
- Extended the capacity-constrained learning framework to online convex and bandit optimization, substantially broadening its scope; resulting work is currently under review at NeurIPS 2026.
- Completed Vector Institute Summer Research Internships (2023, 2024).

Research Assistant

2022 – 2026

University of Toronto, Department of Statistical Sciences

Toronto, ON

- Studied the mathematical foundations of modern machine learning, with projects spanning feature learning in neural networks and high-dimensional statistical estimation.
- Conducted theoretical and large-scale simulation studies of graphical-model estimation.
- Completed MCS Summer Research Internship (2022).

Visiting Researcher

Spring 2023

Sunnybrook Health Sciences Centre

Toronto, ON

- Improved Swin UNETR for 3D segmentation in cancer and musculoskeletal research using dilated attention and a custom loss, outperforming the standard architecture in ablation studies.
- Released the resulting implementation as an [open-source codebase](#) and presented the results at the Sunnybrook Orthopaedic Biomechanics Laboratory seminar.

TECHNICAL SKILLS

Programming: Python, C/C++, Bash.

Machine Learning: PyTorch, JAX, Hugging Face Transformers.

Development Tools: Git, Linux, Slurm, Docker.

WORKSHOPS & TALKS

A Reduction from Delayed to Immediate Feedback for Online Convex Optimization. Poster.

- *Learning Theory Summer School & Workshop* (LTSS 2026), Copenhagen, Denmark (Jun 2026).

Reinforcement Learning with Action-Triggered Observations. Seminar Talk.

- *UofT Statistical Sciences Research Program* (UTSSRP 2026), Toronto, Canada (Jun 2026).

Capacity-Constrained Online Learning with Delays. Poster.

- *Vector AI/ML “Remarkable 2026” Conference*, Toronto, Canada (Feb 2026).
- *Southern Ontario Learning Theory Workshop* (SOLT 2025), Waterloo, Canada (Nov 2025).

ACADEMIC SERVICE & TEACHING

Conference Reviewer

2026 – Present

- International Conference on Learning Representations (ICLR), 2027.
- Conference on Neural Information Processing Systems (NeurIPS), 2026.
- Conference on Learning Theory (COLT), 2026.

Teaching Assistant

Spring 2025

University of Toronto

Toronto, ON

- CSC311, Introduction to Machine Learning.
- CSC363, Computational Complexity and Computability.